

DO NOT WRITE ON TEST



**FINANCIAL MATH &
ANALYSIS CONCEPTS**

~OPEN EVENT~

(190)

STATE 2022

TOTAL POINTS _____ **100 Points**

Test Time: 60 minutes

MULTIPLE CHOICE

Identify the choice that *best* completes the statement or answers the question.

1. Benny is paid \$57,000 a year for his new job. The company Benny works for pays biweekly. How much gross pay does Benny receive in each pay period?
 - a. \$2,192.31
 - b. \$2,375.00
 - c. \$2,280.00
 - d. \$2,111.11
2. Hailey is trying to make a budget. She knows her annual salary is \$75,000. If her company pays twice a month and FICA taxes are 8%, how much will she expect to pay in taxes each check?
 - a. \$230.77
 - b. \$500.00
 - c. \$250.00
 - d. \$460.34
3. Joel worked 36 hours last week. His hourly wage is \$14.80. He has pre-tax deductions of \$100 per week and taxes are 6%. What is his weekly net pay?
 - a. \$124.83
 - b. \$500.83
 - c. \$406.83
 - d. \$312.83
4. Shield Pest Control pays their employees 20% commission on new sales and 5% commission on upgrades to existing clients. One employee sold \$23,544 worth of new contracts and \$10,871 worth of upgrades during the last pay period. How much would this employee's commission be?
 - a. \$4,708.80
 - b. \$6,883.01
 - c. \$5,252.35
 - d. \$1,720.75
5. Leanna works as a sales associate for a regional printer company. Her commission rate is 7.5% for all sales, but as incentive to sell more, the company offers an extra 2% on all sales if the associate hits \$100,000. Leanna sold \$155,782 worth of printers last year, what should her commission be?
 - a. \$14,799.29
 - b. \$11,638.65
 - c. \$13,078.20
 - d. \$13, 683.65

6. If a sales associate in your company sold \$23,450 worth of product and earned a commission check of \$4,221, what would their commission rate be?
 - a. 15%
 - b. 16%
 - c. 18%
 - d. 19%
7. ABC Loans is offering a short-term loan package with a 2.5% rate. If a customer borrows \$2,500 with a payback date of one-year, what amount will they payback?
 - a. \$2,562.50
 - b. \$2,550.00
 - c. \$2,575.00
 - d. \$2,582.50
8. Furnituremart sells furniture on credit with a one-year loan and an interest rate of 3.33%. What would the payback amount be for furniture that cost \$7,500?
 - a. \$7,749.75
 - b. \$7,732.50
 - c. \$7,747.50
 - d. \$7,740.00
9. To help with school, Danielle had to get a loan to buy a computer. She borrowed \$3,000 at 9% for 4 years. How much money will she have to pay back?
 - a. \$4,080.00
 - b. \$3,810.00
 - c. \$4,350.00
 - d. \$4,500.00
10. Staci started up a food truck business. She sells her food in a variety of locations. Here are the most recent locations and the number of meals ordered at each location: Smithfield – 220, Zachary – 288, Weston – 173, and Providence – 191. What is the average number of meals served at the recent locations?
 - a. 97
 - b. 230.5
 - c. 218
 - d. 205.5
11. QwikCalcs is measuring the efficiency of their calculators. For the last year, they had the following percentages of calculators returned due to malfunction: 2%, 3%, 2.75%, 2.5%, 2.5%, 3.125%, 3%, 2.5%, 2%, 2.5%, 1.75%, 2%. Which percentage is the mode?
 - a. 2%
 - b. 2.5%
 - c. 1.75%
 - d. 3.125%

12. Jean is a salesman for a regional tire shop. He sold 1 contract Monday, 2 on Tuesday, 5 on Wednesday, 0 on Thursday, and 2 on Friday. On average, how many contracts did Jean sell daily?
- 2
 - 2.5
 - 3
 - 1.5
13. A company has \$25,000 in assets and \$10,000 in liabilities. What is the ratio of assets to liabilities?
- 2 to 1
 - 5 to 2
 - 2 to 5
 - 5 to 3
14. Company XYZ has about \$3.1 million worth of loans and shareholder's equity of \$13.3 million. What is the debt-to-equity ratio for Company XYZ?
- 4.29
 - 1.06
 - .94
 - .23
15. Burgers and More sells 5 burgers for every 2 chicken sandwiches they sell. If they sold 45 burgers, how many chicken sandwiches did they sell?
- 18
 - 27
 - 23
 - 33
16. A school received a shipment of 500 books. Of the total shipment, 24 of the books were damaged. What percentage of books were damaged?
- 4.4%
 - 4.8%
 - 5%
 - 5.2%
17. Data shows that 60% of customers who rent cars end up returning and renting a car a second time. If a car rental company had 45 return customers, how many total customers did they have?
- 30
 - 60
 - 75
 - 90

18. The price of a company's stock increases from \$32 per share to \$40 per share in a month. By what percent did the stock's price increase?
- 20%
 - 25%
 - 45%
 - 35%
19. A drink company that specializes in juice boxes found that it can lower the cost of their juice boxes by using recycled material. They figure it could lower the cost by 27%. Currently, their cost per juice box is \$.89. What is the new projected cost?
- \$.65
 - \$.69
 - \$.62
 - \$.71
20. Research shows that $\frac{3}{125}$ of all manufacturing is lost due to low quality. If manufacturing produced \$1.923 billion, how much would be lost due to quality concerns?
- \$15.384 million
 - \$46.152 million
 - \$76.92 million
 - \$61.536 million
21. A clothing store did inventory and has 43 red shirts, 26 blue shirts, 22 orange shirts, 80 white shirts, 25 black shirts, 49 green shirts, and 55 yellow shirts. What fraction of the shirts are black and yellow?
- $\frac{1}{5}$
 - $\frac{1}{3}$
 - $\frac{4}{15}$
 - $\frac{2}{15}$
22. Fix'em Right Computers bought \$83,000 in supplies and paid \$36,000 in services (before tax) for normal operating activities. If tax is 9%, how much did they pay in just taxes?
- \$7,470
 - \$3,240
 - \$10,710
 - \$9,850
23. Your company received an invoice for \$10,750 with the terms 3/10 net 30. You pay the invoice 9 days after the invoice date. What is the final payment for the invoice?
- \$10,750.00
 - \$10,642.50
 - \$10,535.00
 - \$10,427.50

24. At Inky Printing, if your purchase is over \$10,000, they offer a discount of \$100. If you purchase \$12,131.25 of products and there is a 6% sales tax, what do you end up paying?
- a. \$12,753.12
 - b. \$12,753.13
 - c. \$12,859.12
 - d. \$12,859.13
25. Opal took her sales team out to lunch after a training. She paid \$271.12 in total including a \$35 tip for the waitress. If sales tax was 7%, how much was the tax?
- a. \$17.74
 - b. \$17.73
 - c. \$15.44
 - d. \$15.45
26. Crazy Shots buys camera accessories for \$22 and sells for them for \$33. How much are the accessories marked up?
- a. 40%
 - b. 33%
 - c. 50%
 - d. 61%
27. Shoes for All Occasions marks up their shoes 35%. If they have a pair of shoes listed for resale at \$115.25, what was the price they paid the wholesaler?
- a. \$40.34
 - b. \$74.91
 - c. \$85.37
 - d. \$86.44
28. An item is purchased at \$5.50 for resale. It is first marked up 65%, then marked down 33%, marked down 15%, marked up 5%. What is the final selling price?
- a. \$5.43
 - b. \$5.17
 - c. \$6.14
 - d. \$5.82
29. A store is selling a sweatshirt for \$25.55. This price is a discount price and it is being advertised as 45% off. What was the original price?
- a. \$37.05
 - b. \$56.78
 - c. \$46.45
 - d. \$39.60

30. You purchased 2,200 bushels of barley at \$6 per bushel, and you expect to have 20% spoilage. If your desired markup is 40% based on cost, what should be your selling price per bushel?
- \$11.75
 - \$9.50
 - \$11.50
 - \$10.50
31. Enchanted Flowers & Gifts purchases 950 poinsettias for Christmas at a cost of \$6.50 per plant. If a 15% spoilage rate is anticipated, at what price should the poinsettias be sold to achieve a 60% markup based on selling price?
- \$12.24
 - \$10.40
 - \$11.58
 - \$10.91
32. Patsy's Produce pays \$0.35 per pound for 500 pounds of oranges. On average, 8% of the oranges will spoil before they sell. Find the selling price per pound needed to obtain a 175% markup on cost.
- \$0.68
 - \$0.98
 - \$1.05
 - \$1.25
33. Papa's Carpentry gained \$375,000 in revenue last quarter with a profit margin of 35.5%. What would be the profit for the quarter?
- \$124,875
 - \$129,000
 - \$133,125
 - \$137,250
34. Taste of Homemade earned \$83,500 in net profit and had a net profit margin of 18.22%. What was the revenue earned by Taste of Homemade?
- \$471,288
 - \$445,788
 - \$463,888
 - \$458,288
35. King's Retail Stores purchased packages of batteries for resale. The first purchase was for 12 units that cost \$5.15 each. The second purchase was 24 units that cost \$4.75. If they sold 24 units, use the FIFO method to calculate the cost of leftover inventory.
- \$61.80
 - \$57.00
 - \$59.40
 - \$58.20

36. Games and More bought a new card game to sell in their store. They bought 25 games at \$12.50, 50 more at \$12.20, and then 10 more at \$12.60. They then sold 15 games. Using the LIFO method, what is the cost of the leftover inventory?
- \$861.50
 - \$861.00
 - \$865.50
 - \$865.00
37. Readers 'R Us purchased coloring books for resale. They bought 70 units at \$13.35, 75 units at \$13.25, and 105 units at \$13.15. They sold 100 units. If they use the weighted average method, how much did the units sold cost Readers 'R Us?
- \$1322.56
 - \$1332.00
 - \$1315.00
 - \$1323.60
38. Games Unlimited purchased a new gaming console system. They had one shipment of 15 consoles at \$500, another of 25 at \$450, and a last one of 10 at \$525. They have sold 20 consoles. Using the weighted average method, what is the cost of inventory left?
- \$14,250
 - \$14,340
 - \$14,400
 - \$14,150
39. Seven years ago, Bakeology purchased an oven for \$2,750. They expect it to have a 20-year life and a salvage value of around \$200. Based on the information provided and using the straight-line depreciation method, what is the net book value of the oven on Bakeology's Balance Sheet?
- \$1857.50
 - \$1787.50
 - \$1927.50
 - \$1350.00
40. Stefen is reviewing the value of the office copy machine. It was purchased at \$1,125 and has a useful life of five years and a salvage value of \$150. Stefen's office has used the copy machine for two years and uses double-declining balance depreciation method. What would be the current net book value of the copy machine?
- \$735
 - \$540
 - \$405
 - \$615

41. Jeremy wants to sell a welder he owns before the net book value drops below \$2,000. He bought the welder for \$4,125 and it has a useful life of six years and a salvage value of \$350. Using the straight-line depreciation method, after how many years should Jeremy sell the welder?
- One year
 - Two years
 - Three years
 - Four years
42. Kitchen Essentials offers cash discounts of 5% off by paying invoices within 15 days, 4% within 30 days, and 3% within 45 days. A customer pays their first invoice of \$1,348.00 in 8 days and their second invoice of \$3,678.00 in 40 days. What did they pay for both invoices combined?
- \$4,861.74
 - \$4,811.48
 - \$4,848.26
 - \$4,875.22
43. You notice that one of your vendors offers a trade discount of 4% if you order over 1,000 quantity and they offer a cash discount of 4% if you pay within 30 days of invoice date. How much would you save if you ordered \$10,000 of product and met both conditions?
- \$800
 - \$784
 - \$878
 - \$924
44. The bank is offering savings account with a 1% simple interest calculated annually. If you opened an account and deposited \$4,537.25, how much would you have in 4 years?
- \$4,582.62
 - \$4,582.80
 - \$4,721.48
 - \$4,718.74
45. The present value of Kay's savings account is \$10,000. What is the future value in one year if the account has 6% interest?
- \$10,000
 - \$10,300
 - \$10,600
 - \$10,900

46. You have \$2,500 in a CD with 2% interest and \$7,500 in another CD with 2.5% interest. Both accounts compound interest quarterly. How much interest would you earn in one year?
- a. \$239.65
 - b. \$237.88
 - c. \$237.50
 - d. \$239.27
47. Michele earns a bi-weekly salary of \$3,594 at the Dragonfly Inn. Next month Michele will be promoted from Assistant Manager to Senior Manager. In her new position she will be paid \$9,345 monthly. How much more per year will Michele earn as a Senior Manager than as an Assistant Manager?
- a. \$18,696
 - b. \$69,012
 - c. \$25,884
 - d. \$7,008
48. Mustang Corp. had \$857 of raw materials at the beginning of the first quarter. During the quarter, \$2,342 of materials were purchased, and \$1,912 of materials were used in production. What is the balance in Raw Materials at the end of the quarter?
- a. \$5,111
 - b. \$1,287
 - c. \$430
 - d. \$427
49. Yukon Inc. sells a fixed asset for 4,500. The asset originally cost \$5,700 and had been depreciated \$1,710 over the course of three years. Will Yukon record a gain or loss on disposal and for how much?
- a. \$1,200 gain
 - b. \$510 gain
 - c. \$1,200 gain
 - d. \$510 loss
50. Brazos Corp. normally pays \$2.84 per ounce of oil. A new supplier is offering a 12-ounce bottle of oil that can be purchased for \$32.84. How much would be saved by purchasing the 12-ounce bottle?
- a. \$1.84
 - b. \$5.04
 - c. \$11.08
 - d. \$1.24